

CY 515: Advanced Inorganic Spectroscopy

Course Code	CY 515
Course Name	Advanced Inorganic Spectroscopy
L-T-P-C	3-0-0-3
Prerequisites	Basic understanding of physical chemistry and preliminary knowledge of quantum mechanics, or with instructor's approval.
Intended for	PG and PhD level
Mutual Exclusion	None
Approval	45th BoA (Feb 2022); 39th Senate (Apr 2023), in the BS / BS-MS Chemical Sciences curriculum

Course Contents

- **Electron Paramagnetic Resonance Spectroscopy:** Origin of EPR signals, g-factor, Presentation of the EPR spectrum, Hyperfine splitting: from protons and from Nuclei $I > 1/2$, Origin of hyperfine interaction, Contributions to the hyperfine interactions in isotropic systems, Anisotropic Effects: Anisotropy in the g value, EPR of triplet states, nuclear quadrupole interaction, line widths, Experimental considerations and application of EPR. (10 hours)
- **Magnetic Circular Dichroism Spectroscopy:** Differences with CD, Theoretical Framework: Definition of MCD Terms, Measurement of MCD Spectra, The Interpretation of MCD Spectra, Case Studies; Diamagnetic Systems (A and B terms) and Paramagnetic Systems (C Terms) Magnetic Vibrational Circular Dichroism (MVCD) and X-Ray magnetic Circular Dichroism (XMCD), Application. (6 hours)
- **Nuclear Quadrupole Resonance Spectroscopy:** Nuclear quadrupole moment, Energies of nuclear quadrupole transitions, Effect of magnetic field on the spectra, Relationship between electric field gradient and molecular structure, Applications, Interpretations of structural information from NQR spectra. (6 hours)
- **Mossbauer Spectroscopy:** The Mossbauer effect, Line width, Recoil energy, Mössbauer active nuclei, Isomer shift, Quadrupole splitting, Magnetic hyperfine interactions, Mossbauer spectroscopy vs. Chemical bond, Structure determination and Analytical applications. (7 hours)
- **Photoelectron Spectroscopy:** Auger electron spectroscopy, Photoionization process, Spin-orbit Coupling, X-ray Photoelectron spectroscopy, Electron Energy Loss Spectroscopy (EELS). (6 hours)
- **X-ray Absorption and Emission Spectroscopy:** Introduction of X-ray absorption spectroscopy (XAS), X-ray emission spectrum (XES), X-ray Absorption spectrum, Theory of X-Ray Absorption Near Edge Structure (XANES) and Extended X-ray Absorption Fine Structure (EXAFS), Application. (7 hours)

Laboratory Modules

None

Text Books

1. D. N. Sathyanarayana., **Handbook of Molecular Spectroscopy: From Radio waves to gamma rays**, I. K. International Publishing House Pvt. Ltd. 2019.
2. Russell S. Drago, **Physical Methods in Inorganic Chemistry**, Van Nostrand Reinhold, 2016.

References

1. Jack D. Graybeal, **Molecular Spectroscopy**, McGraw Hill Education Private Limited, 1988.
2. G. Aruldas, **Molecular Structure and Spectroscopy**, PHI Learning Private Limited, Delhi, 2019.
3. Jeroen A. Van Bokhoven, Carlo Lamberti, **X-Ray Absorption and X-Ray Emission Spectroscopy: Theory and Applications**, John Wiley & Sons, Ltd, 2016.
4. Editors: Edward I. Solomon, A. B. P. Lever, **Inorganic Electronic Structure and Spectroscopy, Volume I: Methodology and Volume II: Application and Case Studies**, John Wiley & Sons, Inc., 2006.
5. Skoog, West and Holler, **Fundamentals of Analytical Chemistry**, Saunders's College publishing, 2013.
6. W. Roy Mason, **A Practical Guide to Magnetic Circular Dichroism Spectroscopy**, Wiley-Interscience, 2007.